

Module 05: Thomas' Write Rule and View Serializability

AKTU BCS501 | Unit 5: Concurrency Control Techniques

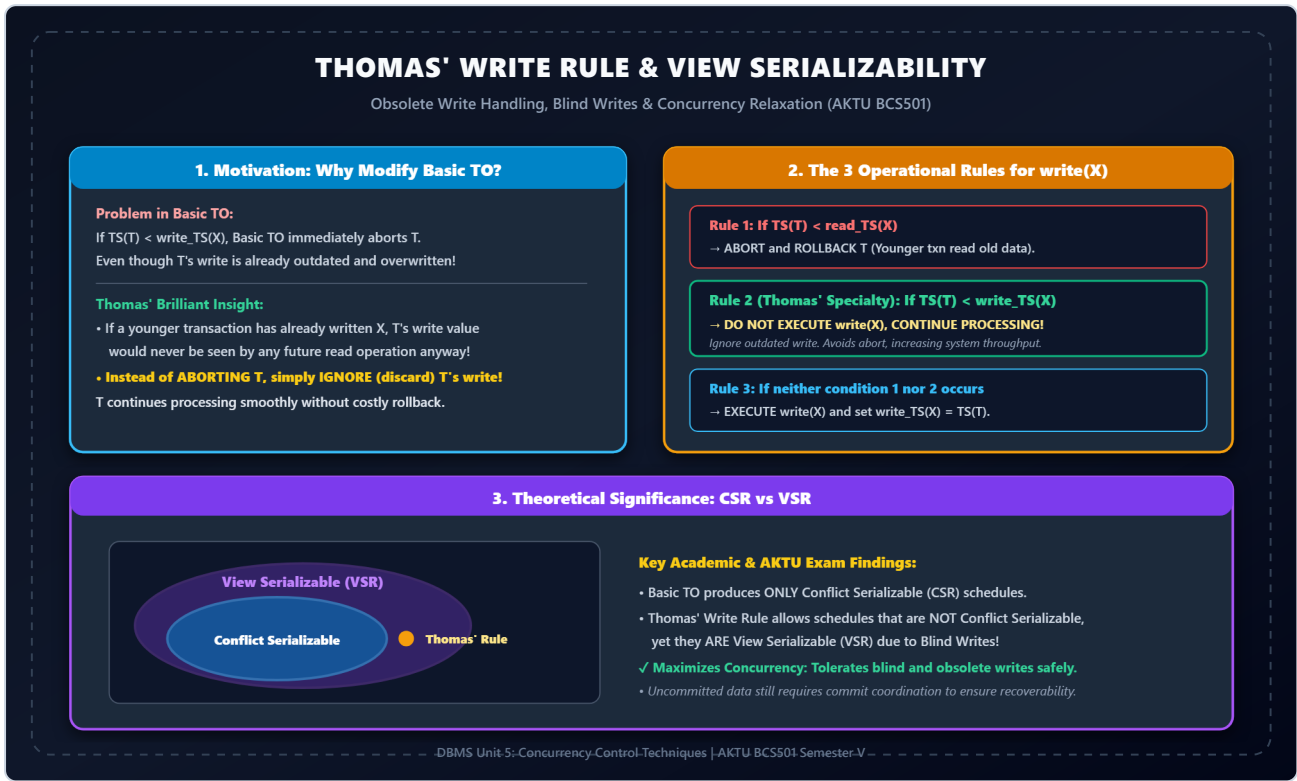


Figure 5.1: Thomas' Write Rule Decision Logic & View Serializability Scope

1. Why Thomas' Write Rule?

In Basic TO, if $TS(T) < write_TS(X)$, T is immediately aborted. Thomas realized that if a younger transaction has already overwritten X, T's write is obsolete. Discarding the write rather than aborting T preserves View Serializability while boosting throughput.

2. The 3 Rules of Thomas' Write Rule

- Rule 1:** If $TS(T) < read_TS(X) \rightarrow$ ABORT T.
- Rule 2 (Thomas' Core):** If $TS(T) < write_TS(X) \rightarrow$ IGNORE write, CONTINUE processing!
- Rule 3:** Otherwise, execute write and set $write_TS(X) = TS(T)$.

View Serializability Invariant

Thomas' Write Rule generates schedules that are strictly **View Serializable (VSR)**, even if they are not Conflict Serializable.